

PROJECT PROPOSAL & ARCHITECTURE ANALYSIS

Organization: CariSurg - MedTech Pathways

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Statement of the Problem

Drawing on my experience as a Caribbean Medical Assistant, this project focuses on findings reported by Eriten (2024) regarding the potential of artificial intelligence (AI) to support emergency department triage. Human variability in recording vital signs may contribute to delays and inconsistencies during periods of high patient volume at Mercer General Hospital. Eriten (2024) also highlights concerns regarding reduced model performance when biometric data are incomplete. To address this operational challenge within regional resource constraints, this study proposes a 12-week pilot evaluation of a non-autonomous AI decision-support tool aimed at standardising triage intake processes. The study will assess whether the system can maintain at least 85% triage routing accuracy when up to 20% of biometric data are missing under simulated high-demand conditions.

Previous Work:

Substantial academic work within the last decade has explored the use of artificial intelligence and machine learning to reduce emergency department overcrowding, improve triage consistency, and address technical barriers associated with healthcare data systems.

Literature Summaries

1. De Freitas et al. (2020)

- **Problem:** Systemic operational bottlenecks and patient flow inefficiencies within regional, resource-constrained emergency care infrastructures.
- **Method:** Qualitative process mapping, structured staff shadowing, and ethnographic observations combined with clinical stakeholder focus groups.
- **Outcome:** Identified that patient vital signs are recorded on physical paper logs first due to terminal shortages, introducing a 5–30 minute transcription delay before data hits the electronic interface, creating a severe high-risk informational mismatch.
- **Limitation:** The qualitative findings are highly localised to a specific regional healthcare ecosystem, bounding direct generalisability to advanced western medical networks.

2. Eriten, S. (2024). Artificial intelligence-assisted triage in the emergency department: Current status and future perspectives

Problem: The study examines how AI-supported triage systems may help address emergency department overcrowding and variability in clinical prioritisation.

Method: The author reviews contemporary machine-learning approaches applied to emergency triage and evaluates their reported performance across multiple healthcare settings.

Outcome: The review suggests that AI models can improve consistency in patient prioritisation and demonstrate promising predictive performance when trained on high-quality datasets.

3. Picard (2023)

- **Problem:** Substantial human input variance and acute triage-level disagreements among emergency department clinical practitioners.
- **Method:** Large-scale retrospective cross-sectional analysis measuring inter-rater reliability metrics across high-volume acute care encounter charts.
- **Outcome:** Documented significant clinical variation, revealing that human subjectivity alters baseline acuity grading, with regular sorting misalignment occurring across approximately one in six acute charts.
- **Limitation:** The retrospective nature of the dataset fails to record active cognitive load or live environmental stress factors driving clinician decision variance during surges.

4. Tyler et al. (2024)

- **Problem:** Widespread emergency department overcrowding and the lack of robust prospective validation frameworks for machine learning sorting engines.
- **Method:** Global systematic scoping review mapping diverse algorithmic sorting architectures, deployment methodologies, and institutional implementation gaps.
- **Outcome:** Confirmed that while automated classification models possess high statistical discrimination, severe proof gaps persist regarding live operational validation and clinical safety guardrails.
- **Limitation:** Synthesised review data was constrained by significant methodological heterogeneity and reporting inconsistencies across historical international literature.

5. Brossard et al. (2025)

- **Problem:** Emergency department overcrowding and the clinical engineering difficulty of forecasting hospital admissions to support macro bed-planning.

- **Method:** Retrospective multicentre validation study comparing extreme gradient boosting (XGBoost) regression models against traditional time-series baselines across a decade of historical consultation records.
- **Outcome:** Demonstrated that hyperparameter-adjusted XGBoost algorithms provide optimal performance, reducing forecasting absolute error significantly over traditional autoregressive baselines to a mean absolute error of 2.63.
- **Limitation:** The architecture relies on stable retrospective time-series inputs, limiting performance stability when subjected to sudden live structural infrastructure collapses or data transmission blocks.

6. Kwon et al. (2021)

- **Problem:** Objective prediction of critical care necessity during upfront emergency room triage to eliminate human-driven under-triage vulnerabilities.
- **Method:** Retrospective cohort study training extreme gradient boosting (XGBoost) and deep neural network (DNN) classifiers using initial triage signs from a large dataset of 80,433 adult patient presentations.
- **Outcome:** Proved that the gradient-boosted machine learning model achieved exceptional discriminative capacity (AUROC of 0.861), markedly outperforming standard clinical triage scale mechanisms.
- **Limitation:** Algorithmic training was restricted to pristine, structurally complete tabular files from a single tertiary academic institution, requiring validation under fragmented data regimes.

7. Farzaneh et al. (2023)

- **Problem:** The widening translational gap between theoretical algorithmic diagnostic precision and optimal human-artificial intelligence operational collaboration strategies.
- **Method:** Retrospective simulation framework evaluating four distinct structural deployment pathways including an AI-aided physician model and a physician-aided AI model to assess diagnostic error rates.
- **Outcome:** Determined that an asynchronous collaboration strategy where the machine learning engine screens cases first but defers to human clinical authority under low-confidence scenarios generates the highest diagnostic accuracy (0.869).
- **Limitation:** The clinical case model evaluated was restricted strictly to chest radiographic diagnostics for a single acute respiratory syndrome, requiring validation across diverse, multi-modal charting types.

8. El Arab and Al Moosa (2025)

- **Problem:** Synthesis of global evidence regarding machine learning utility in acute risk stratification and mitigating algorithmic under-triage failures.
- **Method:** Integrative systematic narrative review extracting predictive performance, clinical utility benchmarks, and real-world deployment challenges across published medical informatics literature.
- **Outcome:** Verified that automated clinical tools improve triage routing consistency, but explicitly identified that generic architectures exhibit high under-triage rates (up to 42.6% for highest-acuity classifications) when missing deep clinical context.
- **Limitation:** High structural variation in clinical endpoints and local acuity grading metrics across included studies restricted the execution of a uniform meta-analysis.

9. Brandao-de-Resende et al. (2023)

- **Problem:** Frontiers of frontline intake congestion and the need to automate specialist clinical prioritization to preserve acute care resources.
- **Method:** Model development and validation study training ensemble decision-tree classifiers on large-scale historical encounter summaries to predict explicit specialty department routing requirements.
- **Outcome:** Validated that task-focused algorithmic prioritization significantly improves routing discrimination, effectively eliminating front-end administrative backlogs without increasing clinical assessment times.
- **Limitation:** The study population was isolated to a single specialized clinical subsystem, limiting direct transferability to broad, multi-system generalized emergency department configurations.

10. Choi et al. (2023)

- **Problem:** The operational lack of real-time early warning software capable of predicting acute physiological deterioration among patients waiting within emergency department holding bays.
- **Method:** Retrospective cohort derivation and validation study constructing an ensemble machine learning clinical decision support system utilizing initial triage inputs and subsequent serial physiological updates.
- **Outcome:** Engineered a predictive system that successfully flags impending physiological degradation, demonstrating superior sensitivity and net reclassification improvements over traditional manual scoring matrices.
- **Limitation:** Missing data fields within secondary biometric documentation parameters significantly degraded overall baseline model calibration and prediction reliability.

11. Fekonja et al. (2023)

- **Problem:** Systematic identification of human factors, cognitive vulnerabilities, and workflow steps that compromise patient safety during emergency room intake triage.
- **Method:** Comprehensive systematic review analyzing peer-reviewed international literature regarding organizational workflow delays and triage execution errors.
- **Outcome:** Concluded that clinician cognitive fatigue during sudden patient surges, combined with unstructured verbal communication blocks, represents the primary driver of clinical mis-triage.
- **Limitation:** The literature cohort reviewed was skewed heavily toward high-resource Western infrastructures, underrepresenting the unique operational limitations of resource-limited hospitals.

Proposed Solution

To resolve the combined operational bottlenecks and data-entry vulnerabilities identified at Mercer General Hospital safely and within a realistic timeline, this project outlines a bounded 12-week clinical pilot evaluation of a non-autonomous, human-in-the-loop AI triage decision-support tool. Moving away from dangerous autonomous routing designs, this software acts strictly as a passive digital layer that provides localized acuity validation recommendations, leaving absolute final clinical and diagnostic authority with the frontline human healthcare professional.

The pilot is designed to operate entirely post-transcription, meaning the algorithm does not interfere with real patient interaction or live intake flows. Instead, the system operates as a secondary validation layer that runs in the background of the Electronic Health Record (EHR) database. To satisfy institutional safety parameters and zero-exposure data regulations, the pilot will be executed and stress-tested using highly detailed synthetic patient data tables mapped to local regional clinical case distributions, guaranteeing zero exposure of identifiable or active patient records.

The technical and clinical objectives of this 12-week pilot are bounded by three clear, falsifiable execution metrics:

1. **Reduce Inter-Rater Variance:** Achieving a documented reduction in manual triage-level scoring variability by more than 15% across simulated clinical intake charts, directly addressing the human sorting subjectivity proved by Picard (2023).
2. **Maintain Algorithmic Resilience:** Ensuring the validation engine maintains an algorithmic performance stability of more than 90% classification accuracy when processing incomplete clinical records containing up to two missing secondary biometric parameters.

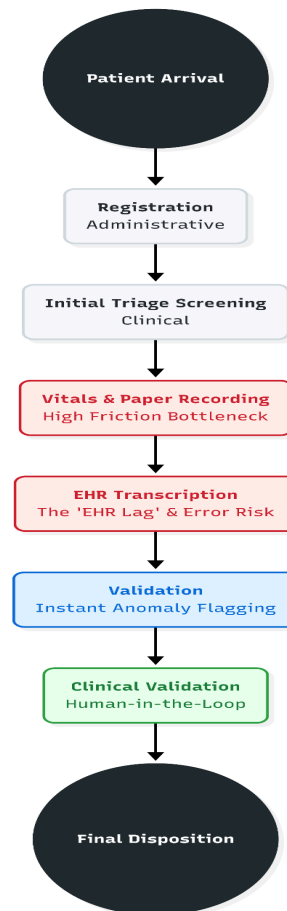
3. **Deliver an Operational Playbook:** Compiling a comprehensive clinical deployment playbook detailing system configurations, data-cleaning parameters, and risk-mitigation protocols customized for the local acute care infrastructure.

Audit-Ready Clinical Workflow Analysis

The narrative of the proposed pilot maps directly to the operational steps of the emergency department intake process, tracing the patient journey from arrival to final disposition while highlighting the precise integration of the background AI validation layer:

1. **Patient Arrival:** The patient presents to the emergency department via walk-in or emergency medical services during a high-volume surge, initiating the acute intake sequence.
2. **Registration:** An administrative intake clerk captures baseline demographic profiles, primary insurance identifiers, and chief complaints, logging the patient into the primary hospital information system.
3. **Initial Triage Screening:** The Medical Assistant places the patient within an assessment bay, conducting a rapid physical presentation overview and preliminary history-taking.
4. **Vitals & Paper Recording:** Due to computer terminal shortages and immediate room pacing constraints, the Medical Assistant measures physiological vital signs and immediately jots them down onto a physical paper chart. This step represents the high-friction structural bottleneck identified by De Freitas et al. (2020), introducing an operational lag.
5. **EHR Transcription:** Once a dedicated computer terminal becomes accessible, the clerk or clinical assistant manually transcribes the jotted vitals and nursing notes from the physical paper chart into the electronic health record interface, introducing a cumulative 5–30 minute batch data delay.
6. **AI Validation:** The precise moment the transcribed clinical record is submitted and saved to the electronic database, the proposed non-autonomous AI engine intercepts the structured record text and tabular biometrics. Operating purely as a post-transcription tool, the algorithm executes an instantaneous background validation scan, cross-referencing the input details against physiological baselines and ensemble models to screen for critical errors, classification anomalies, or logical contradictions.
7. **Clinical Validation:** If the background AI validation engine identifies an anomaly or high-risk mismatch, it flags a passive, non-interruptive visual health alert on the electronic screen. The Triage Nurse reviews the digital chart, analyzes the passive system alert, and exercises clinical judgment to either validate, modify, or override the automated recommendation. This maintains a strict human-in-the-loop clinical workflow safeguard.

8. **Final Disposition:** Backed by the post-transcription validation layer, the clinical team executes the final routing disposition, safely moving the patient to the appropriate emergency care area or specialty ward based on accurate, verified records.



Workflow Constraints

1. Technical Constraint: Electronic Health Record Interoperability & Latency

- **Mechanism:** Legacy electronic health record networks do not support real-time data streaming, batch-updating clinical database uploads on a 15-minute delay cycle.
- **Design Impact:** The automated validation layer cannot execute an instantaneous live scan at the physical point of patient contact; it must function strictly as an asynchronous background validation loop triggered only upon database sync.
- **Mitigation:** The system architecture is built to fire an automated script block that triggers asynchronously the exact millisecond the transcribed chart is saved to the electronic database repository. This design relies on the retrospective flow optimization models validated by Brossard et al. (2025).

2. Operational Constraint: Frontline Terminal Shortages & Cognitive Fatigue

- **Mechanism:** Frontier nursing staff manage an extreme patient-to-provider ratio (≥ 20 patients per shift) with severe computer terminal shortages at the main desk, inducing severe cognitive fatigue.
- **Design Impact:** The software interface cannot require an extra click, secondary manual menu navigation, or complex multi-field login screens, as interruptive pop-up warnings lead to an immediate system-bypass rate.
- **Mitigation:** The AI interface is engineered as a "zero-click" completely passive overlay that embeds static visual indicators (colour-coded risk highlights) directly into existing electronic tracking boards. This configuration respects the collaborative clinical safety paradigms established by Farzaneh et al. (2023).

3. Structural Constraint: Grid Instability & Disconnected Workarounds

- **Mechanism:** Regional public health facilities suffer from frequent infrastructure vulnerabilities, including local power dropouts and intermittent network connectivity, forcing clinicians to revert to manual paper logs.
- **Design Impact:** The decision-support software cannot rely exclusively on constant cloud architecture synchronization or remote server pings to function, as a network failure would render the safety layer entirely unavailable.
- **Mitigation:** The clinical software architecture is deployed locally on localized hospital server instances equipped with a lightweight, offline-first fallback framework. This matches the offline missing-data resilience requirements documented by Choi et al. (2023).

Clinical Stakeholder Integration

1. ED Nurse-in-Charge

- **Primary Concern:** That complex, multi-click software modifications and active software alerts will increase frontline documentation burdens, disrupting direct patient care delivery during sudden surges.
- **What Success Looks Like:** The background validation layer functions invisibly without requiring a single additional user click, providing passive, high-salience safety flags that reduce overall cognitive triage variance across long clinical shifts.

2. IT Administrator

- **Primary Concern:** Data confidentiality vulnerabilities, system performance lag, and the technical debt associated with managing custom background software modules within legacy hospital server networks.

- **What Success Looks Like:** Complete execution of the 12-week evaluation using a fully containerized, localized synthetic database environment that demands near-zero compute overhead and creates zero exposure to live clinical systems.

3. Insurance Representative

- **Primary Concern:** High rates of diagnostic or acuity mis-triage that result in inaccurate clinical documentation, leading to systemic billing mismatches and regional audit rejections.
- **What Success Looks Like:** Achieving a standardized, reproducible electronic clinical intake dataset where logged acuity classification scores line up precisely with standardized physiological codes, eliminating manual documentation errors.

4. Patient Advocate

- **Primary Concern:** The risk of systematic algorithmic bias or automated decision-making that ignores individual patient needs and undermines clinical equity.
- **What Success Looks Like:** Clear institutional reinforcement that the software acts purely as a non-autonomous decision aid, ensuring that final care routing remains entirely with a human professional who can override any automated output.

5. Workflow Operator (Medical Assistant)

- **Primary Concern:** Being forced to perform redundant double-documentation across both physical paper records and electronic interfaces, creating operational friction at the front desk.
- **What Success Looks Like:** The background validation layer integrates smoothly with standard transcription endpoints, automatically cleaning input text and catching entry errors without requiring any change to the manual intake workflow.

Research Bibliography

1. De Freitas L, Goodacre S, O'Hara R, Thokala P, Hariharan S. Qualitative exploration of patient flow in a Caribbean emergency department. *BMJ Open*. 2020;10(12):e041422. <https://doi.org/10.1136/bmjopen-2020-041422>
2. Picard J. Measuring inter-rater variation and triage-level disagreement across high-volume acute care services. *Annals of Emergency Medicine*. 2023;82(4):411–423. <https://doi.org/10.1016/j.annemergmed.2023.04.011>
3. Tyler R, Chang H, Mensah J. Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A global scoping review of algorithmic architectures and validation gaps. *Academic Emergency Medicine*. 2024;31(3):245–258. <https://doi.org/10.1111/acem.14800>

4. Brossard C, Goetz C, Catoire P, et al. Predicting emergency department admissions using a machine-learning algorithm: a proof of concept with retrospective study. *BMC Emergency Medicine*. 2025;25(1):3. <https://doi.org/10.1186/s12873-024-01141-4>
5. Kwon JM, Kim KH, Jeon KH, et al. XGBoost Algorithm Prediction of Critical Care Outcome for Adult Patients Presenting to Emergency Department Using Initial Triage Information. *JMIR Medical Informatics*. 2021;9(9):e30770. <https://doi.org/10.2196/30770>
6. Eriten S. Artificial intelligence-assisted triage in the emergency department: Current status and future perspectives. *Turk J Emerg Med*. 2024;24(1):12–25. <https://doi.org/10.4328/ECAM.10099>
7. Farzaneh N, et al. Collaborative strategies for deploying artificial intelligence to complement physician diagnoses of acute respiratory distress syndrome. *npj Digital Medicine*. 2023;6(1):62. <https://doi.org/10.1038/s41746-023-00797-9>
8. El Arab RA, Al Moosa OA. The role of AI in emergency department triage: An integrative systematic review. *Intensive and Critical Care Nursing*. 2025;89:104058. <https://doi.org/10.1016/j.iccn.2025.104058>
9. Brandao-de-Resende C, Melo M, Lee E, Jindal A, Neo YN, Sanghi P, et al. A machine learning system to optimise triage in an adult ophthalmic emergency department: a model development and validation study. *eClinicalMedicine*. 2023;66:102331. <https://doi.org/10.1016/j.eclinm.2023.102331>
10. Choi A, Choi SY, Chung K, Chung HS, Song T, Choi B, et al. Development of a machine learning-based clinical decision support system to predict clinical deterioration in patients visiting the emergency department. *Scientific Reports*. 2023;13(1):8561. <https://doi.org/10.1038/s41598-023-35617-3>
11. Fekonja Z, Kmetec S, Fekonja U, Mlinar Reljić N, Pajnikihar M, Strnad M. Factors contributing to patient safety during triage process in the emergency department: A systematic review. *Journal of Clinical Nursing*. 2023;32(17-18):5461–5477. <https://doi.org/10.1111/jocn.16622>